

Show all of your work in the exam booklet provided to you. Full credit will not be awarded for correct answers unless sufficient supporting work is presented.

1. (15 pts) The matrix $A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$ reduces to $\begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

a) Find a basis for the column space of A.

b) Find a basis for the null space of A.

c) Is the vector $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ in the column space of A?

2. (9 pts) Find the coordinate vector for $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ with respect to each basis below.

a) $B_1 = \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$ b) $B_2 = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ c) $B_3 = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

3. (12 pts) Let A be any $m \times n$ matrix. Decide if each statement is true or false and give a short explanation for your answer.

- a) AA^T is an $m \times m$ matrix.
- b) The diagonal entries of AA^T cannot be negative.
- c) If $m < n$, then the null space for A must be infinite.
- d) $\det(AA^T) = \det(A)\det(A^T)$.

4. (6 pts) Find the LU factorization of $A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix}$.

5. (10 pts) Let A be any $m \times n$ matrix. Prove that the null space of A is a subspace of $\mathbb{R}^n$.

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6. (16 pts) Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and define $\text{Trace}(A) = a + d$.

a) (9 pts) Prove that Trace is a linear transformation from the vector space of all 2×2 matrices to $\mathbb{R}$.

b) (3 pts) Give three examples of matrices in the kernel of the Trace mapping?

c) (2 pts) Is the Trace mapping one-to-one?

d) (2pts) Is the Trace mapping onto?

7. (10 pts) For which values of λ is $\det \begin{bmatrix} 1-\lambda & 0 & 0 \\ 2 & 4-\lambda & 0 \\ 3 & 5 & 6+\lambda \end{bmatrix} = 0$?

8. (8 pts) Use cofactor expansion to compute $\det \begin{bmatrix} 3 & 4 & 2 & -2 \\ 0 & 5 & 0 & -4 \\ 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & -1 \end{bmatrix}$

9. (8 pts) Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 3 & 2 & 1 \\ -1 & -2 & 1 \\ 0 & 0 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 3 \\ 2 & -1 \end{bmatrix}$.

a) What is the size of AB ?

b) Write the first column of AB as a linear combination of the columns of A .

c) Write the second row of AB as a linear combination of rows of B .

d) Use column-row multiplication to write AB as a sum of matrices.

10, (6pts) Let A be an $n \times n$ matrix. Which of the following statements are equivalent to saying A is invertible? List all correct answers.

a) $\det(A) = 0$.

b) The columns of A are linearly independent.

c) The columns of A^T are linearly independent.

d) The rows of A are linearly independent.

e) The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

f) The linear transformation $T(\mathbf{x}) = A\mathbf{x}$ is one-to-one.